

Worked Solution for 2018 H1 Chem Prelim

1	2	3	4	5	6	7	8	9	10
B	B	B	A	A	B	C	D	C	B
11	12	13	14	15	16	17	18	19	20
B	C	B	C	C	D	B	C	D	A
21	22	23	24	25	26	27	28	29	30
D	A	B	A	D	B	B	C	A	C

- 1 *Use of the Data Booklet is relevant to this question.*

Which sample of gas contains the same number of atoms as 2 g of hydrogen gas?

A 22 g of carbon dioxide

B 12 g of steam

C 10 g of neon

D 8 g of methane

Amount of $\text{H}_2 = 2/2 = 1 \text{ mol}$

Number of moles of H atoms = 2

Amount of $\text{CO}_2 \text{ gas} = 22/44 = 0.5 \text{ mol}$

Number of moles of atoms of $\text{CO}_2 = 0.5 \times 3 = 1.5 \text{ mol}$

amount of $\text{H}_2\text{O} = 12/18 = 2/3 \text{ mol}$

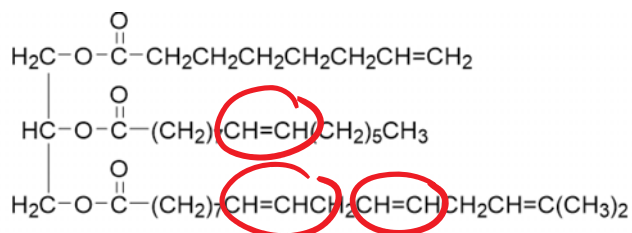
Number of moles atoms from $\text{H}_2\text{O} = 2/3 \times 3 = 2 \text{ mol}$

Amount of Ne gas = Number of moles of atoms = $10/20 = 0.5 \text{ mol}$

Amount of methane = $8/16 = 0.5 \text{ mol}$

Number of moles of atoms from methane = $0.5 \times 5 = 2.5 \text{ mol}$

- 2 The diagram shows the structure of a tri-ester.

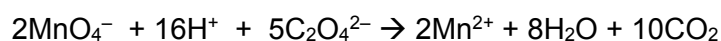


What is the total number of cis-trans isomers possible for this tri-ester?

Criteria of cis-trans; C=C and different atoms/grps attached to the C in C=C.

- A 2^1 B 2^3 C 2^4 D 2^5

- 3 Ethanedioate ions, $\text{C}_2\text{O}_4^{2-}$, are oxidised by acidified aqueous potassium manganate(VII) to give carbon dioxide according to the following equation:



What volume of $0.020 \text{ mol dm}^{-3}$ potassium manganate(VII) is required to completely oxidise $1.0 \times 10^{-3} \text{ mol}$ of the salt $\text{KHC}_2\text{O}_4 \cdot \text{H}_2\text{C}_2\text{O}_4$?

- A 20 cm^3 B 40 cm^3
 C 80 cm^3 D 100 cm^3

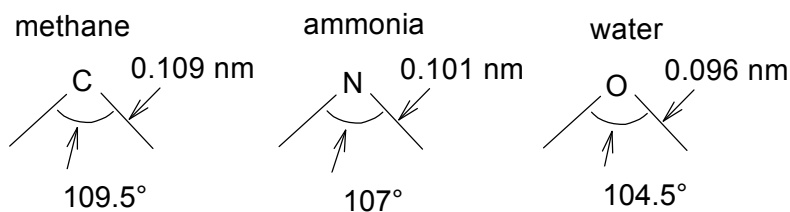
Mole ratio of $\text{MnO}_4^- : \text{C}_2\text{O}_4^{2-} = 2:5$

Amount of $\text{C}_2\text{O}_4^{2-} = 2(1.0 \times 10^{-3})$
 $= 2.0 \times 10^{-3} \text{ mol}$

Amount of $\text{MnO}_4^- = 2/5 \times 2.0 \times 10^{-3}$
 $= 0.0008 \text{ mol}$

Volume of $\text{MnO}_4^- = 0.0008 / 0.020$
 $= 40 \text{ cm}^3$

- 4 The bond lengths and bond angles in the molecules of methane, ammonia and water may be represented as follow:



What causes this trend in the bond angles shown?

- 1 Lone pair–bond pair repulsion is greater than bond pair–bond pair repulsion.
- 2 Decreasing repulsion between the bond pair of electrons as the electronegativity of the central atom increases.
- 3 Increasing repulsion between hydrogen atoms as the bond length decreases.

Bond angle in a molecule depends on the number of bond pair and lone pair of electrons. By VSEPR theory, lone pair–lone pair repulsion > lone pair–bond pair repulsion > bond pair–bond pair repulsion.

Methane: 4 bond pairs around C

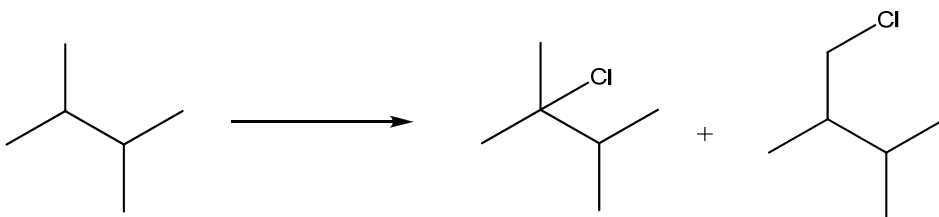
Ammonia: 3 bond pairs, 1 lone pair around N

Water: 2 bond pairs, 2 lone pairs around O

Hence, option 1 is correct.

- A** 1 only
- B** 1 and 2 only
- C** 2 and 3 only
- D** 1, 2 and 3

- 5 2,3-Dimethylbutane can undergo free radical substitution with chlorine gas. Ignoring any stereoisomers, how many different mono-chlorinated organic products are formed when it is heated with limited chlorine gas?



- A** 2 **B** 3 **C** 4 **D** 6

6 Which statement about Group 1 metals is correct?

A Melting point increases down the group.

False: strength of metallic bond depends on the size of cation and no of delocalised electrons. Down the grp, ionic radius increases, hence weaker the attraction b/w sea of delocalised electrons and cation. Thus weaker the metallic bond strength.

B Reducing strength increases down the group.

Down the grp, size of cation increases hence weaker the attraction of valence electrons to the nucleus. Thus, it is easier to lose electron down the group → stronger the reducing agent.

C Each ground state atom contains a pair of electrons in its valence shell.

False: The atom contain an electron

D The second electron is more difficult to be removed than the first electron, as M^+ has a noble gas configuration.

False: it is more difficult to lose the 2nd electron as a lot of energy is required to remove the inner quantum shell electrons since it is nearer to the nucleus, which experience strong attraction.

7 An enzyme may be described as being specific in its activity.
What is meant by this this statement?

A The enzyme will denature at high temperature.

B The enzyme helps to increase the speed of a chemical reaction.

C The enzyme will catalyse a narrow range of chemical reactions.

D The enzyme is a biological catalyst which remains chemically unchanged at the end of the reaction.

8 Chlorate (I) ion undergoes disproportionation to give chloride ion and another chlorine containing product.

Which of the following could be the chlorine-containing product?

Disproportionation: same element is oxidised and reduce at the same time.

Chlorate (I) ion, Cl = +1

Chloride, Cl = -1

Hence the other Cl shd have a O.S higher than +1

A Cl_2

O.S of Cl = zero

B Cl_2O

O.S of Cl = +1

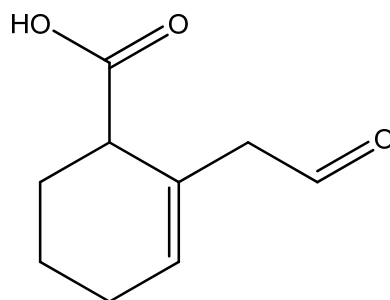
C ClO^-

O.S of Cl = $-1 + 2 = +1$

D ClO_3^-

O.S of Cl = $-1 + 6 = +5$

- 9 Which are the products formed when compound **X** reacts with the following reducing agents?

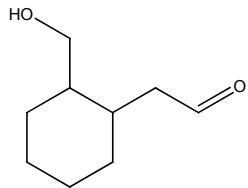
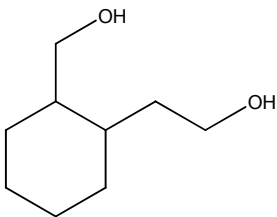
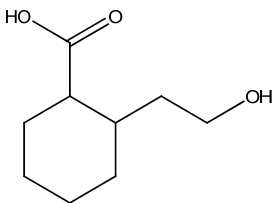
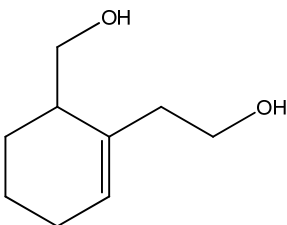
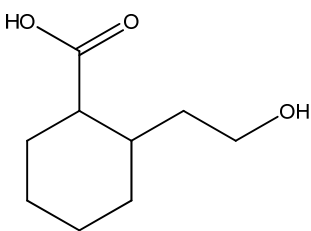
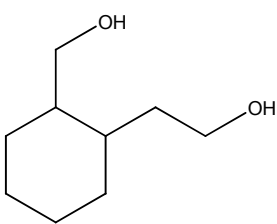


compound **X**

H_2 , Pt: reduce alkenes $\rightarrow$ alkanes and aldehyde $\rightarrow$ primary alcohol

LiAlH_4 in dry ether: reduce carboxylic acid $\rightarrow$ primary alcohol and aldehyde $\rightarrow$ primary alcohol

	H_2 , Pt	LiAlH_4 in dry ether
A		

B		
C		
D		

10 A substance **Y** has the following properties.

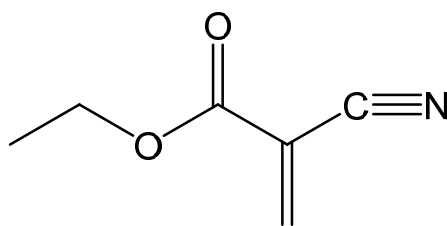
- melting point – 39 °C
- boiling point 357 °C
- conduct electricity when solid and molten

What is the structure of **Y**?

Since **X** can conduct electricity when solid and molten it must be metallic or giant molecular (graphite). All giant molecular cpd will have high melting and boiling point.

- A** ionic
- B** metallic
- C** giant molecular
- D** simple molecular

11 Ethyl cyanoacrylate, $C_6H_7O_2N$, is the major component of superglue.



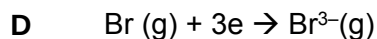
ethyl cyanoacrylate

Which statements about the molecule are true?

- 1 It contains 5 lone pairs of electrons.
True: 2 lone pairs on each of oxygen atom and 1 lone pair of electrons on N.
 - 2 It reacts with Br₂ to give a pair of cis-trans isomers.
False: no cis-trans in the product due to the absence of C=C
 - 3 It contains 4 π bonds.
True: 1 π on C=O and C=C and 2 π on CN.
 - 4 It can be hydrolysed by dilute NaOH to give a carboxylic acid and an alcohol.
False: carboxylate salt is formed due to alkaline condition.
- A 1 only
- B 1 and 3 only**
- C 2 and 3 only
- D 3 and 4

12 Which equation represents the third ionisation energy of bromine?

- A $\text{Br (g)} \rightarrow \text{Br}^{3+}(\text{g}) + 3\text{e}$
- B $\text{Br}^{2-}(\text{g}) + \text{e} \rightarrow \text{Br}^{3-}(\text{g})$
- C $\text{Br}^{2+}(\text{g}) \rightarrow \text{Br}^{3+}(\text{g}) + \text{e}$**



- 13** When 20 cm³ of 0.5 mol dm⁻³ NaOH(aq) is neutralised with an equal volume of 0.5 mol dm⁻³ HNO₃(aq), the temperature of the mixture rose by 5.2 °C.

What would be the temperature change if 40 cm³ of 0.25 mol dm⁻³ NaOH(aq) is reacted with an equal volume of 0.25 mol dm⁻³ HNO₃(aq)?

Magnitude of temperature rise depends on moles of water produced and the total volume of solution.

Temperature increases with no of moles of water produced.

Temperature decreases with volume of solution. since $q = mc\Delta T$

Experiment 1: mole of water produced = 0.01 mol

Total volume = 40 cm³

Experiment 2: mole of water produced = 0.01 mol

Total volume = 80 cm³

Since volume is doubles as compared to experiment 1, ΔT will be reduced by half;
 $5.2/2 = 2.6\text{ °C}$

A 1.3 °C **B** 2.6 °C **C** 5.2 °C **D** 10.4 °C

- 14** Iodine–131 is a radioactive isotope with a half-life of 10 days. Given that radioactive decay is a first-order reaction, what fraction of the isotope would remain after 50 days?

No of half- life = $50/10 = 5$

$1 \rightarrow 1/2 \rightarrow 1/4 \rightarrow 1/8 \rightarrow 1/16 \rightarrow 1/32$

A $\frac{1}{10}$ **B** $\frac{1}{20}$ **C** $\frac{1}{32}$ **D** $\frac{1}{50}$

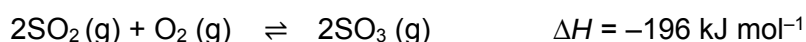
- 15 The molar mass of ethanoic acid is 120.0 g mol^{-1} when dissolved in ether but 60.0 g mol^{-1} when dissolved in water.

Which one of the following statements best explains the phenomenon?

Mr x 2 → dimer is only formed (via hydrogen bonding) when ethanoic acid is dissolved in ether.

- A Ethanoic acid is more soluble in ether than in water.
- B Ethanoic acid molecules form strong covalent bonds with ether molecules.
- C Ethanoic acid molecules form extensive hydrogen bonds with water molecules.
- D Ethanoic acid molecules form dative bonds with another ethanoic acid molecule when dissolved in ether.

- 16 The enthalpy change of reaction for the conversion of sulfur dioxide to sulfur trioxide is shown below.



Which of the following statements is correct?

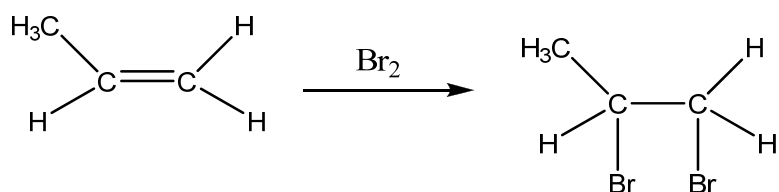
- A When the volume of the vessel is compressed, the yield of SO_3 decreases.
False: Pressure is increases. POE shifts right where there are fewer gaseous particles.
- B When a catalyst is introduced to the system, the yield of SO_3 increases.
False: Catalyst only increases the rate of reaction.
- C When the temperature is increased, only the rate of the backward reaction is increased.
False: rate of backward and forward will increases with temperature but to different extent.
- D When the temperature is increased, K_c decreases.

True: temperature is increased, POE shifts left to favour endothermic reaction to absorb excess heat. Rate of backward reaction increases more than forward. Hence K_c decreases.

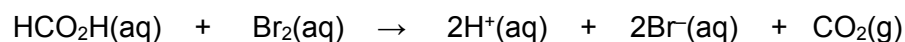
- 17 Bromine has two isotopes, ^{79}Br and ^{81}Br .

Propene can react with bromine liquid in the dark to form organic products with different relative molecular mass.

How many different relative molecular mass values does the organic products have?

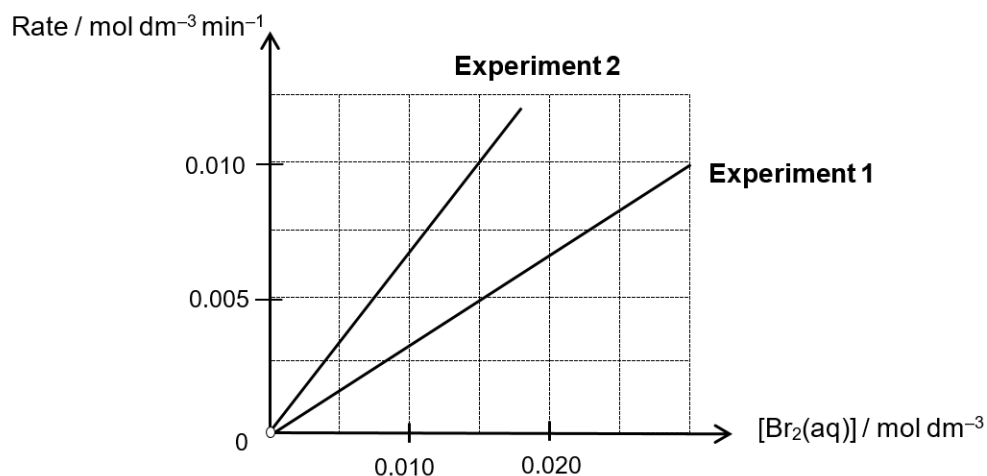


- A 2
- B 3 ($^{79}\text{Br}^{81}\text{Br}$, $^{79}\text{Br}^{79}\text{Br}$ or $^{81}\text{Br}^{81}\text{Br}$)**
- C 4
- D 5
- 18 Two separate experiments were conducted to study the kinetics of the following reaction in the presence of excess methanoic acid.



Experiment	Initial $[\text{HCO}_2\text{H}] / \text{mol dm}^{-3}$
1	0.8
2	1.6

The experimental results were represented in the graph as shown.



Which of the following is the correct rate equation for above the reaction?

Straight line, rate is directly proportional to [Br₂]. 1st order w.r.t to [Br]

Comparing 2 experiments,

When [Br₂] is the same in both experiments, the rate increases by 2x when [HCO₂H] increase by 2 times. Hence it is 1st order reaction w.r.t to [HCO₂H]

- A Rate = $k [\text{Br}_2]$
- B Rate = $k [\text{HCO}_2\text{H}]$
- C Rate = $k [\text{HCO}_2\text{H}][\text{Br}_2]$**
- D Rate = $k [\text{HCO}_2\text{H}]^2[\text{Br}_2]$

- 19 The chloride of X and oxide of Y respectively are separately mixed with water. The two resulting mixtures have different effect on litmus.

Which of the following represents the correct elements?

AlCl₃, MgCl₂, SiCl₄ and PCl₅ hydrolyses in water to give H⁺

SO₃ and P₄O₁₀ are acidic oxide

MgO and NaO basic oxide

	X	Y
A	Al	S

B	Mg	P
C	Si	P
D	P	Na

- 20 A pure sample of $\text{NH}_3(\text{g})$ is introduced into a vessel of constant volume and an equilibrium is established as shown below.



The value of the final pressure is then found to be 25 % greater than if only NH_3 were present. What is the mole fraction of N_2 in the reaction mixture?

- A 0.10
B 0.25
C 0.45
D 0.75

	$2\text{NH}_3(\text{g}) \leftrightarrow \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$		
Initial/ mol	1	0	0
Change/mol	-2x	+x	+3x
Equilibrium/mol	1-2x	x	3x

Pressure $\propto$ moles since P_{t} is found to be 25% more at equilibrium as compared to initial; this implies that the total no of moles is $1 \times 125\% = 1.25 \text{ mol}$

$$1.25 = 1 - 2x + x + 3x$$

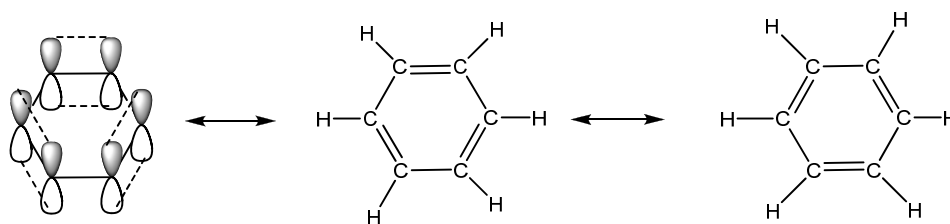
$$0.25 = 2x$$

$$0.125 = x$$

$$\text{Mole fraction of } \text{N}_2 = x/(1+2x) = 0.125/1.25 = 0.10$$

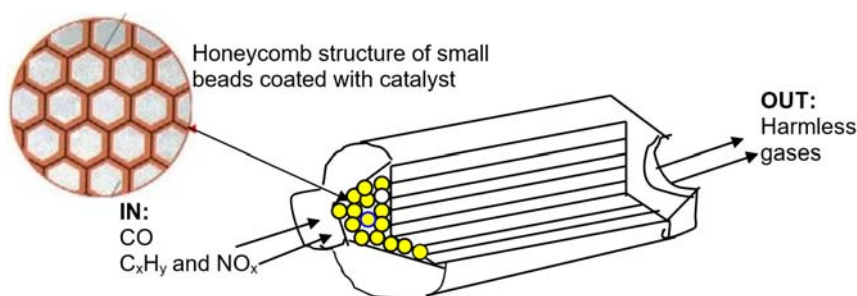
- 21 What is the correct description of the bonding between the two carbon atoms in benzene?

Benzene, C_6H_6



- A two pi bonds that are longer than a C—C single bond.
 - B two pi bonds that are shorter than a C—C single bond.
 - C a sigma and a pi bond that are longer than a C—C single bond.
 - D a sigma and a pi bond that are shorter than a C—C single bond.**
- (C-C bond in benzene is like a 1.5 C=C bond)

- 22 Catalytic converters in car exhaust system helps to convert harmful emission gases into harmless ones. The following diagram shows the interior of a catalytic converter.



Which of the following statement is correct?

- A The forces of attraction holding the carbon monoxide, unburnt hydrocarbons and oxides of nitrogen onto the surface of the catalyst are stronger than the forces of attraction holding on the product molecule onto the surface.**
 - B The catalyst coated on the honeycomb surface provides a small (shd be large) surface area to volume ratio to increase the rate of reaction.
 - C Homogenous (shd be heterogenous) catalyst is used in the catalytic converted to lower the activation energy for conversion of harmful gases
 - D Nitrogen dioxide gas (N_2 gas) is one of the products of the reactions.
- 23 What is the electronic configuration of copper in Cu_2O ?

O.S of Cu in Cu_2O = +1

Cu = $[Ar] 3d^{10}4s^1$



24 Which of the following statements about the Period 3 elements sodium to chlorine, and their compounds, is correct?

1 The ionic radius decreases across the period.

Size of cation and anion decreases across the period. However, the ionic of all cations is smaller than the anions across the period.

2 The oxide of the elements become more acidic across the period.

Basic oxide $\rightarrow$ amphoteric oxide $\rightarrow$ acidic oxide

3 The boiling points of the elements decreases in the order phosphorus, sulfur, chlorine.

Boiling point of $\text{P}_4 < \text{S}_8$ (highest boiling point due to larger electron cloud $\rightarrow$ stronger id-id)

Boiling point of $\text{S}_8 > \text{Cl}_2$

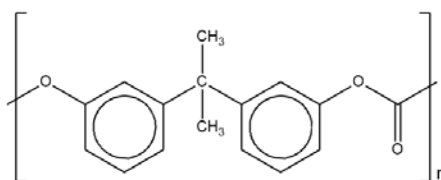
A 2 only

B 2 and 3 only

C 1 and 2 only

D 1,2 and 3

25 Polycarbonates (PC) are polymers containing carbonate groups in their chemical structures. Some major smartphone manufacturers such as Samsung and Apple started using PC in their phones casing.



PC

Which row correctly describes PC?

	type of polymer	reaction of polymer with ethanoic acid
A	addition	no reaction
B	condensation	no reaction
C	addition	hydrolysis
D	condensation	hydrolysis

Ester bond in the polymer chain → condensation polymerisation which can hydrolyse in acidic medium (vinegar solution is ethanoic acid)

26 Which reaction will give propanoic acid as one of the products?

A heating $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ under immediate distillation with potassium dichromate.

Aldehyde is formed

B heating $\text{CH}_3\text{CH}_2\text{CONHCH}_3$ under reflux with dilute aqueous sulfuric acid.

Acidic hydrolysis → $\text{CH}_3\text{CH}_2\text{COOH} + \text{CH}_3\text{NH}_3^+$

C heating $\text{CH}_3\text{CH}_2\text{OCOCH}_3$ under reflux with dilute aqueous sulfuric acid.

Acidic hydrolysis → $\text{CH}_3\text{CH}_2\text{OH} + \text{CH}_3\text{COOH}$

D heating $\text{CH}_3(\text{CH}_2)_2\text{COOCH}_3$ under reflux with dilute aqueous sulfuric acid.

$\text{CH}_3(\text{CH}_2)_2\text{COOCH}_3 \rightarrow \text{CH}_3(\text{CH}_2)_2\text{COOH} + \text{CH}_3\text{OH}$

27 What is the maximum size, in at least one dimension, of a nanomaterial?

1-100 nm → $100 \times 10^{-9} = 10^{-7} \text{ m}$

A $1 \times 10^{-6} \text{ m}$

B $1 \times 10^{-7} \text{ m}$

C $1 \times 10^{-8} \text{ m}$

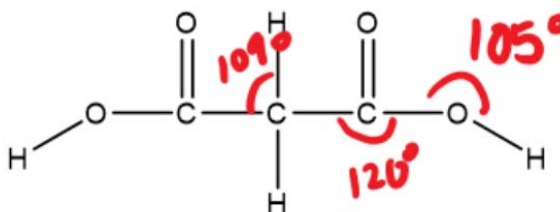
D $1 \times 10^{-9} \text{ m}$

28 The two most common isotopes of titanium are $^{56}_{22}\text{Ti}$ and $^{58}_{22}\text{Ti}$.

Which statement about the isotopes of titanium is correct?

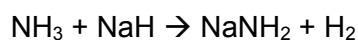
- A** Both isotopes have the same numbers of neutrons.
neutrons = mass no – proton no
- B** One of the isotopes has more protons than the other.
- C** The electronic configuration of Ti^{2+} ion for both isotopes is the same.
Electronic configuration depends on number of electrons.
- D** In the same electric field strength, $^{58}_{22}\text{Ti}^+$ will be deflected more than $^{56}_{22}\text{Ti}^+$.
Angle of deflection $\propto$ charge/mass

- 29** The diagram shows the structure of the malonic acid.



Which of the following shows the correct set of the bond angles present in the malonic acid?

- A** 105° , 109° , 120°
- B** 109° , 180° only
- C** 120° , 180° only
- D** 105° , 109° only
- 30** Ammonia gas can react with sodium hydride in the following equation:



Which statement correctly describes the behaviour of ammonia in this reaction?

- A** It acts as an Arrhenius acid.
- B** It acts as an Arrhenius base.
- C** It acts as Bronsted-Lowry acid.

NH_3 donated a H^+ $\rightarrow$ NH_2^-

D It acts as Bronsted-Lowry base.

END OF PAPER